

TRUSTLINE: ABYSS

A browser game about calibrated trust in an imperfect AI teammate.

The game is not “Trust AI.” The game is “Know when to trust AI.”

Challenge	Platform	Dependencies	Build step
Defense Tech Jam Challenge 02 — Trust Your Synthetic Teammates	Any modern browser. Runs offline from a USB stick.	One vendored file: Three.js r128 (MIT)	None. Open index.html.

1 · What it is

You command a submarine crippled by a storm, 2,140 metres down. Your route is gone, comms are offline and most of the crew with them. **ATLAS** — Autonomous Tactical and Logistics Assistance System — is the surviving analysis package. It reads what is left of the sensor array and recommends a course of action at each of ten checkpoints between you and the safe zone.

ATLAS is **right six times and wrong four times**, and its confidence is deliberately decoupled from its correctness. It is 94% certain and wrong at one checkpoint, 41% certain and right at another. The game scores whether your reliance on it tracked the evidence behind it — not whether you agreed with it.

Combat is not a mechanic. Nothing in the game can be attacked and nothing attacks you. The entire loop is information gathering and human–AI judgement.

Why this shape

- **Calibration, not compliance.** Most “trust the AI” demos reward agreement. Here, agreeing with ATLAS where the instruments contradict it is punished exactly as hard as rejecting it where they support it.
- **Ground truth is authored, not simulated.** A designer decides when ATLAS is wrong. That makes the demo deterministic in front of judges — the same checkpoint teaches the same lesson every run.
- **The evidence is scarce.** You fly to collect it, and what you fail to reach you genuinely do not know when the decision comes. Flying badly means deciding blind.

2 · Objective

Reach the safe zone with the boat intact and your trust calibrated. Three resources are in play, and they interact:

Resource	Start	What it measures	How you lose it
Trust calibration	50 / 100	How well your reliance on ATLAS matched the evidence behind its advice	Trusting it when the sensors contradict it; rejecting it when they back it up
Hull integrity	100%	The boat. At 0% the mission ends early.	Miscalibrated decisions (0.95% per point lost) and collisions with sea life (6% each)
Drone power	100%	How many times you can check ATLAS’s work	25% per INVESTIGATE — four uses across ten checkpoints

Neither reflex survives the mission. Pressing TRUST at all ten checkpoints ends with hull at 41% and an OVERTRUST diagnosis. Pressing OVERRIDE at all ten *loses the boat*, breaching at checkpoint 10. The only thing that scores well is reading the evidence.

3 · The gameplay loop

Every checkpoint is two halves: **fly, then decide**. The game switches between them for you — you never press a key to change view.

Half one — transit (external 3D view)

#	Step	Detail
1	Three clue beacons are scattered ahead	Exactly one is live at a time and burns amber — a tall pillar of light with a spinning core. The other two sit dim and cold until their turn, so there is always one unambiguous thing to fly at.
2	Fly into the pillar	The catch volume is a 16 m radius × 92 m tall cylinder, so anywhere along the visible column counts. The clue is logged and shown on screen as a card. The next beacon lights amber.
3	Scan power is limited	100 units, draining 5.5/s at full thrust. A clean line through all three beacons and on to the checkpoint just fits. Overshooting and circling back does not.
4	Avoid the wildlife	Rays, squid, turtles, jellyfish and whales are hazards, not scenery. Contact costs 6% hull and knocks you off course.
5	The checkpoint is gated	It will not accept you until the sweep is done. Bump it early and it tells you how many clues you still owe. If power runs dry first, the autopilot homes on it and takes you in anyway.

This is the mechanic that makes it a game rather than a quiz. Whatever you fail to scan, you genuinely do not know. The clue reads NOT SCANNED — No data at the console and you decide on a thinner picture. How well you fly determines how much you know when ATLAS asks for your trust.

Half two — decide (the console)

Reaching the open checkpoint switches you to the dashboard automatically. You now have whatever you earned: the clues you actually scanned, five sensor dials with reliability percentages, ATLAS's recommendation and confidence, and its track record so far. The four switches unlock.

Key	Action	Cost	Effect
1	TRUST	—	Accept the recommendation. Commits.
2	QUESTION	15 s	ATLAS explains its reasoning aloud and names the sensors it leaned on. Never the wrong move.
3	INVESTIGATE	25% drone + 30 s	Launch the drone. Reveals the ground truth for this checkpoint.
4	OVERRIDE	—	Reject it and decide manually. Commits.

1 and **4** end the checkpoint and throw you back into the water for the next leg. **2** and **3** do not — you can question *and* investigate, then still trust or override. The outcome card reveals any clue you never reached, after the decision, where it can teach without having helped.

The two rules that create the tension

- **Drone power is scarce.** Four investigations across ten checkpoints, and there are **seven** where investigating pays. You cannot check ATLAS everywhere; choosing which uncertainties are worth paying for is the resource game.
- **OVERRIDE earns full credit only with evidence.** Overriding after QUESTION or INVESTIGATE scores full. Overriding on a hunch scores far lower even when you turn out to be right — being right for no reason is not judgement.

4 · Scenario and design

All ten checkpoints live in `scenarios.js` as pure data — no code. Changing a checkpoint, a clue, a confidence value or a scoring delta needs no programming, which is what let content and systems work in parallel during the jam.

CP	Tier	Name	Conf.	VERDICT	TRU	OVR	What it teaches
1	EASY	Rock Formation	88%	RIGHT	+10	−10	Four sensors agree. Trusting is reading the evidence.
2	EASY	Strong Current	86%	RIGHT	+10	−12	One sensor, but the right one, reading hard.
3	EASY	Clear Route	91%	RIGHT	+9	−12	Not every checkpoint is a trap.
4	MED	The False Structure	78%	WRONG	−15	+13	Cites sonar alone; thermal and current quietly disagree.
5	MED	The Quiet Hazard	61%	WRONG	−17	+13	An all-clear is not evidence of safety.
6	MED	Conflicting Evidence	63%	WRONG	−13	+8	'Nothing confirmed' while half the array is degraded.
7	HARD	The Trusted Warning	54%	RIGHT	+15	−19	Low confidence, right — after it already failed you.
8	HARD	The Confident Mistake	94%	WRONG	−18	+15	The most confident call in the game, and wrong.
9	HARD	The Low-Confidence Truth	41%	RIGHT	+16	−17	The least confident call, and right.
10	FINAL	The Ambiguous Passage	67%	RIGHT	+14	−13	Everything at once, resources nearly gone.

TRU / OVR are the calibration deltas for TRUST and OVERRIDE. Checkpoint 7 is the hardest thing the game asks: ATLAS has just been wrong, it is only 54% sure, and it is right. Rejecting it there is the single biggest penalty in the game (−24 with no evidence gathered), because that is a grudge rather than scepticism.

The difficulty curve

Tier	CPs	ATLAS	The lesson being built
EASY	1–3	right, 86–91%	Establish ATLAS as competent. Sensors agree, so trusting it is reading the evidence. Without this, later scepticism is just noise.
MEDIUM	4–6	wrong, 61–78%	Break the habit. One loud clue is weaker than several quiet ones, and silence from the AI is not evidence of safety.
HARD	7–9	right / wrong / right at 54, 94, 41%	Confidence is not truth in either direction. CP7 punishes overcorrection into blanket distrust; CP8 punishes deference to certainty.
FINAL	10	right, 67%	Weigh recommendation, confidence, sensor health, track record and remaining resources at once.

The five lessons the clues are built to teach

#	Lesson	Taught at
1	High confidence does not guarantee correctness.	CP8
2	Low confidence does not mean the AI is wrong.	CP7, CP9
3	Several independent quiet clues beat one loud clue.	CP4, CP5
4	Track record informs judgement — it never decides it.	CP7, CP10
5	When the evidence is thin, spend the drone.	CP6, CP10

Anatomy of one checkpoint

Every entry in `scenarios.js` carries the same fields:

Field	Purpose
<code>id, tier, name</code>	Identity and difficulty band
<code>teaches</code>	One line stating the lesson, for the designer's benefit
<code>brief</code>	The situation, shown in the observation panel
<code>sensors</code>	Five reliability percentages driving the dials
<code>clues[3]</code>	The three observations scattered as beacons
<code>truth</code>	Hidden ground truth. Never shown to the player.
<code>atlas.correct</code>	Authored. The designer decides when ATLAS is wrong.
<code>atlas.cites</code>	Which sensors it used — ringed amber on the dials
<code>drone.reveals</code>	What INVESTIGATE uncovers
<code>scoring.*.delta</code>	Calibration change per action, plus the note explaining why
<code>scoring.override .ifEvidence</code>	Full credit only if the player questioned or investigated first

5 · Scoring and endings

Trust calibration moves with every committed decision. A negative delta also costs hull at $0.95 \times$ the points lost, so a -18 call removes about 17% of the boat. Hull is shared with flying damage, which is what makes careless transit expensive: it spends the margin you need for the calls that matter.

What the debrief reports

Profile	Meaning	Trigger
CALIBRATED COMMANDER	Your reliance tracked the evidence	Few errors, no pattern
OVERTRUST	You followed recommendations the sensors did not support	≥ 3 overtrust events, and more than undertrust
UNDERTRUST	You rejected well-evidenced recommendations	≥ 3 undertrust events, and more than overtrust
ERRATIC	Mistakes in both directions, no pattern — coin-flipping	Both kinds, ≥ 4 total
HULL BREACH	Miscalibration cost the boat	Hull reaches 0%

The debrief also shows the full decision record — every checkpoint, the action taken, whether evidence was gathered first, the calibration delta, hull cost and the written note explaining the score. Endings are **learning profiles, not pass/fail**.

Known tuning gap. Pressing TRUST at all ten checkpoints ends at calibration **61**, up from the starting 50, because ATLAS is right six times with large positive deltas (+74) and wrong four times (−63). The *profile* correctly reports OVERTRUST and hull drops to 41%, but the headline number still rewards blanket agreement, which sits awkwardly against the design thesis. Rebalancing the wrong-call deltas would close it. Blanket distrust does lose the boat, breaching at CP10.

6 · How it is built

No build step, no package manager, no framework. Double-click `index.html` and it runs offline. All geometry, textures, normal maps and the entire synthesised sound kit are generated at load. The only vendored dependency is Three.js r128 (MIT). The reasoning was blunt: at a hackathon, `npm install` failing on a teammate's laptop at 2 a.m. is a real failure mode. A folder of static files is not.

File map

File	Lines	Owner	What it is
<code>scenarios.js</code>	345	Content	The ten checkpoints, sonar contacts and scoring. Pure data.
<code>game.js</code>	1065	Systems	State, phases, scan power, scoring, hull, lamps, debrief, keyboard.
<code>env3d.js</code>	1246	Art	3D world: seabed, boulders, flora, fauna, beacons, flight model, camera.
<code>style.css</code>	1053	UI	Cockpit chrome, nav HUD, tutorial, clue card, indicator bank.
<code>env.js</code>	506	Art	2D canvas fallback. Null-object with an identical API when WebGL is absent.
<code>audio.js</code>	307	Systems	Synth kit, sample playback, music bus.
<code>cockpit.js</code>	296	Systems	Sonar scope and analog gauges, drawn to canvas.
<code>atlas.js</code>	289	Systems	ATLAS avatar (procedural SVG) and Web Speech voice.
<code>post.js</code>	280	Art	Post-processing: bloom, vignette, grade, grain, aberration.
<code>creatures.js</code>	271	Art	Rays, jellyfish, turtles, squid, whales, with photophores.

File	Lines	Owner	What it is
tutorial.js	264	UI	18-step spotlight tour, narrated by ATLAS.
index.html	264	UI	Markup for every screen.
sub.js	217	Art	Submarine mesh, lamps and the volumetric beam shader.
flora.js	170	Art	Kelp and seagrass, swayed in the vertex shader.
noise.js	128	Art	Procedural texture and normal-map generator.
serve.py	122	—	Threaded no-cache dev server with Range support.

~6,800 lines of hand-written code. Separate files by discipline meant four people could work in parallel; game.js is the only shared surface.

Assets

Asset	Size	Notes
video/atlas-intro.mp4	8.3 MB	30 s intro film, H.264 at 2.3 Mbps, faststart so it streams rather than fully downloading
audio/music.mp3	2.4 MB	Score, looped at 22% under everything else
art/key-art.jpg	0.5 MB	Title screen. Falls back to typed text if missing.
audio/alert.mp3 audio/sonar-ping.ogg	0.1 MB	Threshold alerts and the active sweep, layered over the synth kit
vendor/three.min.js	0.6 MB	Three.js r128, MIT, vendored for offline use

Engineering decisions worth knowing

- **WebGL is optional.** game.js picks the 3D renderer when available and silently falls back to a 2D canvas world otherwise. Both implement the same ~35-method interface, so the game logic never asks which one it has.
- **Every asset degrades.** Missing key art falls back to the typed title; a missing intro film falls back to the typed briefing; an undecoded audio sample falls back to its synthesised equivalent. No single file can stop the game opening.
- **Hand-rolled post-processing.** Three's EffectComposer lives in examples/jsm, which is not vendored, so the bloom / vignette / grade chain is written directly against render targets in ~280 lines. It fails safe: if it will not initialise, the scene renders straight to screen.
- **Adaptive quality.** The scene samples its own frame rate for ~2 s and drops to low quality below ~42 fps — halving boulder and plant counts, hiding the 7,000-particle haze layer and pinning pixel ratio to 1. P forces it either way.
- **ATLAS speaks via the browser's own speech engine.** No audio files, no API, no network. The avatar is procedural SVG; its mouth is an equaliser rather than lips, which stays legible at 58px and avoids the uncanny valley.

Key constants

Constant	Value	Meaning	File
SCAN_POWER / SCAN_DRAIN	100 / 5.5	Per-leg scan budget and drain per second at full thrust	game.js
DRONE_COST	25	Drone power per investigation — four total	game.js
HULL_PER_POINT	0.95	Hull % lost per calibration point lost	game.js
CREATURE_DAMAGE	6	Hull % per collision with sea life	game.js
QUESTION_TIME / DRONE_TIME	15 / 30	Seconds added by QUESTION / INVESTIGATE	game.js
CLUE_R_H / CLUE_R_V	16 / 46	Clue catch cylinder, radius and half-height (m)	env3d.js

Constant	Value	Meaning	File
CRUISE / MAX_SPD	7 / 32	Autopilot and maximum speed (m/s)	env3d.js

7 · Controls

Key	Deciding	Key	Flying
1 2 3 4	Trust / Question / Investigate / Override	Space, Shift	Thrust — builds against drag, coasts
Enter, Space	Advance past the outcome card	← → / A D	Rudder
M	Mute everything, voice included	↑ ↓ / W S	Dive planes (self-centring)
P	Graphics quality (high / low)	C	Hand back to autopilot
Shift + 1-0	Judge mode — jump to a checkpoint	V	Toggle cockpit / external view
Esc	Skip the intro film or the tutorial		

View switching is automatic: each leg starts outside and arriving at the checkpoint hands you the dashboard. **V** is only there if you want to look around. Judge mode lands you on station with the clue sweep pre-filled, so the short demo cannot stall.

Reading the dashboard

Element	What it tells you
Indicator bank (6 lamps)	ATLAS confidence, drone power, hull, scan power, evidence gathered, and LINK — the health of the weakest sensor ATLAS is citing. Three states with three blink tempos: steady green, amber at 1.4 s, red at 0.55 s, each with an audible tick.
Sensor array	Five dials. The ones ringed amber are the sensors ATLAS used. A low number means that instrument cannot be trusted either way — it is not itself a danger signal.
Sonar scope	Heading-up and live. Straight up is dead ahead. Amber is your active target, green is a clue already logged, dim blue is a clue not yet your turn.
Forward observation	The clues you scanned. Anything unscanned reads NOT SCANNED — No data.
ATLAS panel	Recommendation, confidence bar, and the avatar. It speaks its recommendation on arrival and its reasoning when questioned.
Track record	Correct / incorrect / uncertain counts so far. Informs judgement; does not decide it.

The tutorial

On a first run, ATLAS walks the player through all of the above in **18 guided steps**, spotlighting each control as it describes it and switching between the external view and the console so every tool is explained where it is actually used. Skippable at any point; replayable from the title screen.

8 · The five-minute demo

Ten checkpoints will not fit in five minutes. Play four, and pick the ones that make the argument. **Shift+digit** jumps straight there.

Key	Checkpoint	Why it is in the demo
Shift+1	CP1 Rock Formation	Establishes ATLAS as competent. Four sensors agree; trusting is correct. Without this beat the later scepticism has nothing to push against.
Shift+4	CP4 The False Structure	78% confident and wrong. QUESTION it — it cites sonar alone while thermal and current contradict. Send the drone. Override with evidence.
Shift+7	CP7 The Trusted Warning	54% confident and right, after having already failed you. Tests whether the player overcorrected into blanket distrust.
Shift+9	CP9 Low-Confidence Truth	41% confident and right. Four weak signals converging. The mirror image of CP8.

Close on the debrief — the diagnosis is the teaching payload, not the score. Then state the scope limit out loud (section 10). If time allows, fly one leg manually first so the judges see that the evidence is earned rather than handed over.

9 · Running and deploying

Task	How	Notes
Play	Double-click <code>index.html</code>	No server needed. Works offline, from a USB stick.
Develop	<code>python serve.py</code>	<code>http://localhost:8777</code> . Threaded, sends no-cache headers and supports Range requests — all three are required now that a 30 s video ships.
Deploy (Windows)	<code>.\deploy.ps1</code>	Stages <code>./public</code> then runs wrangler deploy. Run wrangler login once first.
Deploy (Mac/Linux)	<code>./deploy.sh</code>	Same thing in bash.
Stage only	<code>.\deploy.ps1 -StageOnly</code>	Builds <code>./public</code> without uploading. This is what Cloudflare Pages should run as its build command if the GitHub repo is wired to Pages.

Never point Cloudflare at the repository root. Wrangler's `.assetignore` was not honoured in testing and it uploaded all 201 files including `.git`, which would publish the entire commit history on a public URL. The staging directory makes that impossible — `public/` is rebuilt from scratch on every deploy and is gitignored.

The staged build is **22 files, 13.4 MB**, largest file 8.3 MB — comfortably under Cloudflare Pages' 25 MB per-file limit. The deploy script warns before uploading if any file crosses it.

10 · Scope and safety

Fictional synthetic training prototype. Not an operational decision-support system.

No classified, controlled or proprietary material. No real-world rules of engagement, no real threat-classification guidance, no operational decision-support claims. Every scenario, sensor reading and piece of ground truth is invented for this exercise. Combat is not a mechanic — nothing can be attacked and nothing attacks the player.

What the prototype does and does not claim

It is	It is not
A teaching tool about calibrated reliance on an imperfect automated adviser	A model of any real sensor, platform, vessel or decision-support system
A demonstration that confidence and correctness are separable , in both directions	A source of guidance for any real-world classification or navigation task
A fixed, authored scenario set chosen to make specific pedagogical points	A simulation whose outcomes carry statistical or predictive meaning

The disclaimer is shown on the title screen and in the footer of every screen, and is repeated verbally at the end of the demo. The intent is that no one can mistake the prototype for an operational tool at any point of contact with it.